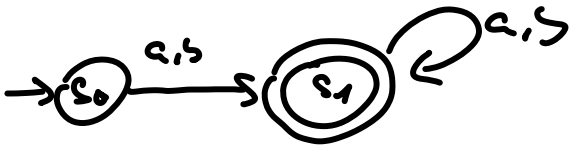
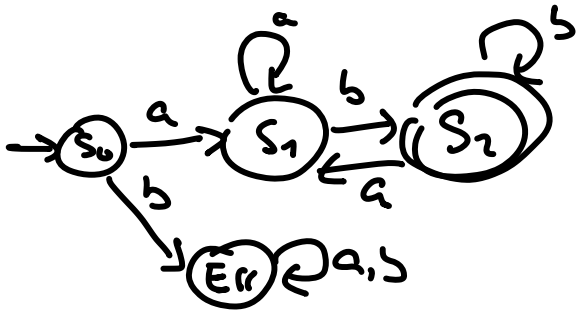
L - DEAs

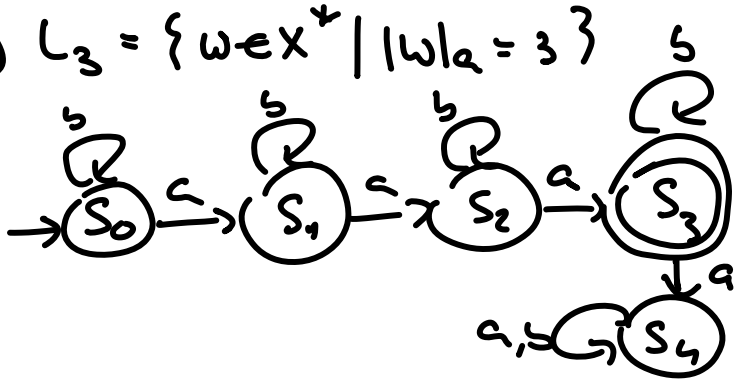
1) a) $L_1 = X^+$



b) $L_2 = \{ w \in X^* \mid w = aXb,\ X = \{a,b\}^* \}$

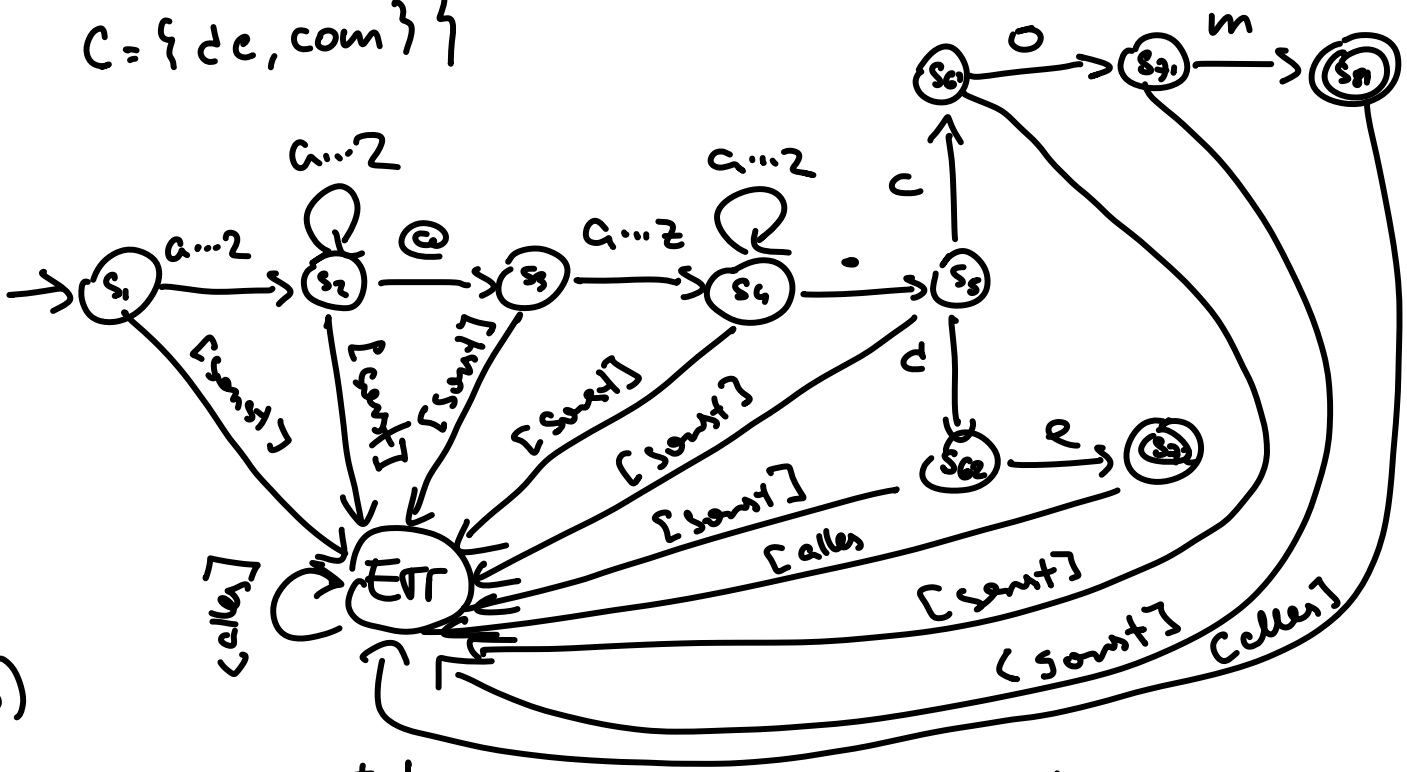


c) $L_3 = \{ w \in X^* \mid |w|_a = 3 \}$



2)

$L = \{ w \in X^* \mid w = A@B.C,\ A = \{a,b,\dots,y,z\}^+,\ B = \{a,b,\dots,y,z\}^+,$
$C = \{de, com\} \}$



3)

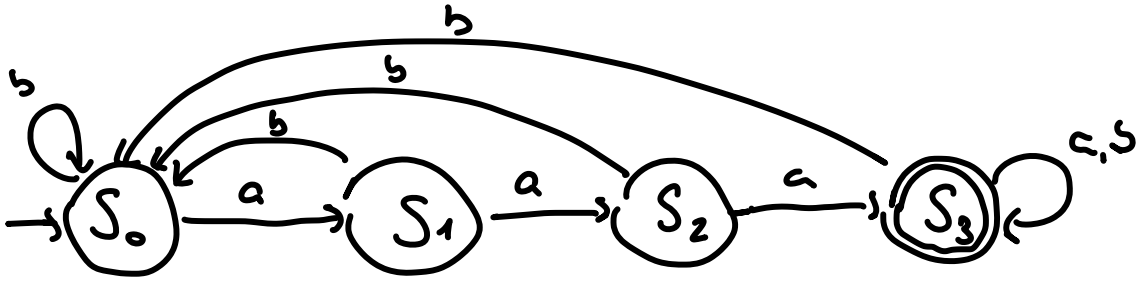
$L(A) = \{ w \in X^+ \mid w_i \neq w_{i+1},\ 1 \leq i < |w|,\ i \in \mathbb{R} \}$

oder

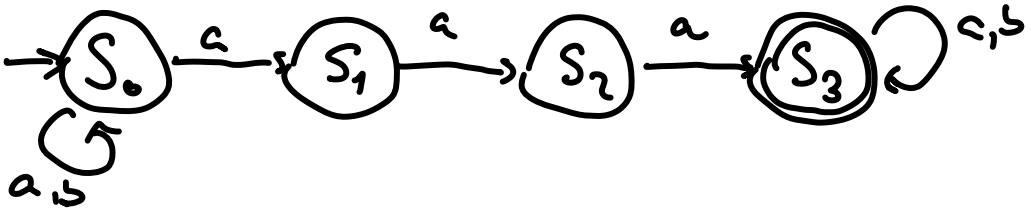
$L(A) = \{ \{ 1(01)^n,\ 0(10)^n \mid n \geq 0 \} \vee \{ (01)^n, (10)^n \mid n \geq 1 \},\ n \in \mathbb{R}_0 \}$

2-NEAs

1) $L = \{ w \in X^* \mid w = y a^5 z : y, z \in X^* \}$



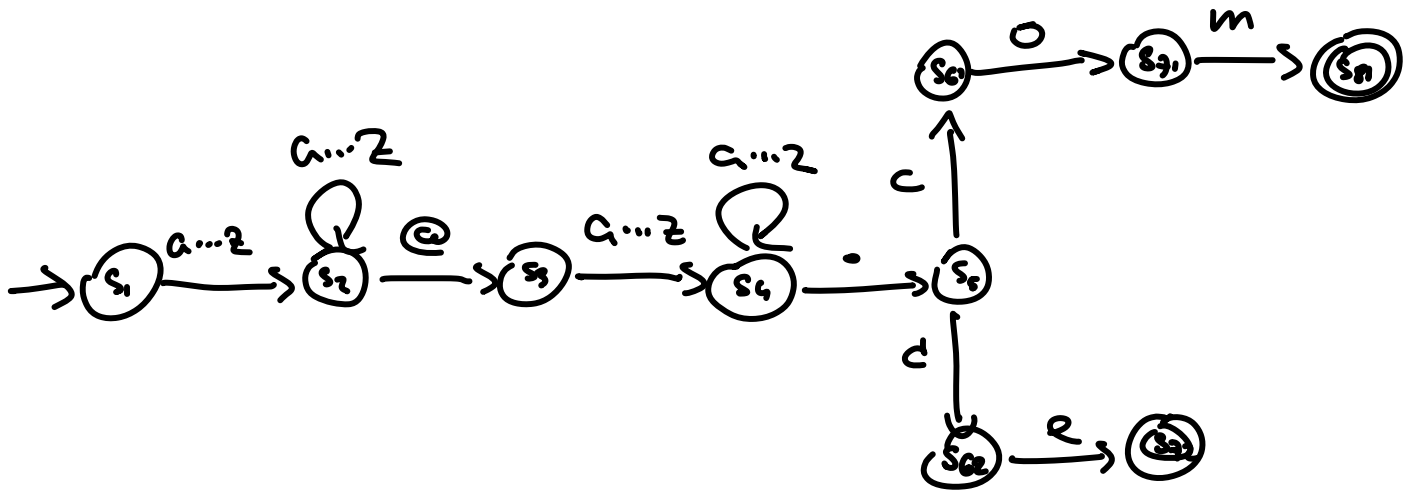
δ	a	b
→ S_0	S_1	S_0
S_1	S_2	S_0
S_2	S_3	S_0
◎ S_3	S_3	S_3



δ	a	b
→ S_0	$\{S_0, S_1\}$	$\{S_0\}$
S_1	$\{S_2\}$	$\emptyset$
S_2	$\{S_3\}$	$\emptyset$
◎ S_3	$\{S_3\}$	$\{S_3\}$

2)

$L = \{ w \in X^* \mid w = A@B.C,\ A = \{a,b,\dots,y,z\}^+,\ B = \{a,b,\dots,y,z\}^+,$
$C = \{de, com\} \}$

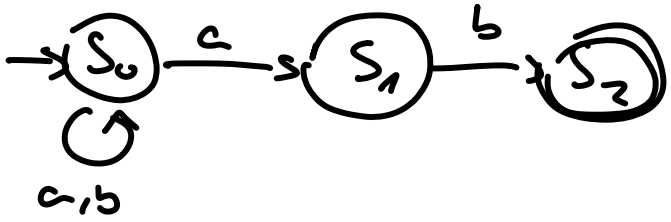


δ	$x \in X \setminus \{c,o,m,d,e,.,@\}$	c	o	m	d	e	.	@
S_1	$\{S_2\}$	$\{S_2\}$	$\{S_2\}$	$\{S_2\}$	$\{S_2\}$	$\{S_2\}$	$\emptyset$	$\emptyset$
S_2	$\{S_2\}$	$\{S_2\}$	$\{S_2\}$	$\{S_2\}$	$\{S_2\}$	$\{S_2\}$	$\emptyset$	$\{S_3\}$
S_3	$\{S_4\}$	$\{S_4\}$	$\{S_4\}$	$\{S_4\}$	$\{S_4\}$	$\{S_4\}$	$\emptyset$	$\emptyset$
S_4	$\{S_4\}$	$\{S_4\}$	$\{S_4\}$	$\{S_4\}$	$\{S_4\}$	$\{S_4\}$	$\{S_5\}$	$\emptyset$
S_5	$\emptyset$	$\{S_{61}\}$	$\emptyset$	$\emptyset$	$\{S_{61}\}$	$\emptyset$	$\emptyset$	$\emptyset$
S_{61}	$\emptyset$	$\emptyset$	$\{S_{71}\}$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
S_{71}	$\emptyset$	$\emptyset$	$\emptyset$	$\{S_{81}\}$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
S_{81}	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
S_{62}	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\{S_{72}\}$	$\emptyset$	$\emptyset$	$\emptyset$
S_{72}	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

3 Aufg. DEAs

1) a) $L_1 = \{ w \in \{0,1\}^* \mid w = x\,ab,\ x \in X^* \}$

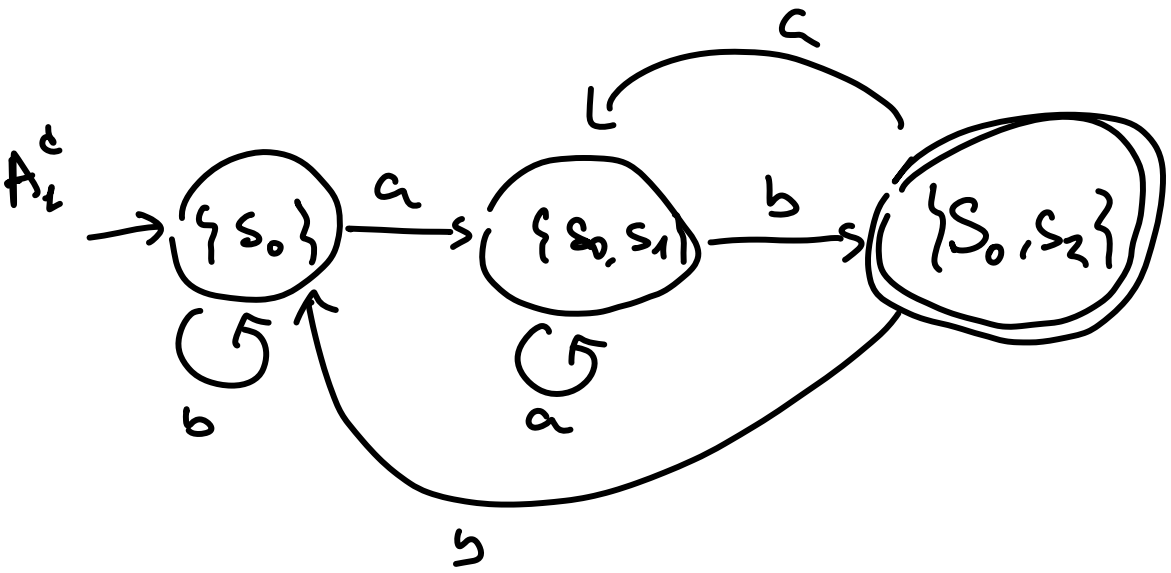
A1



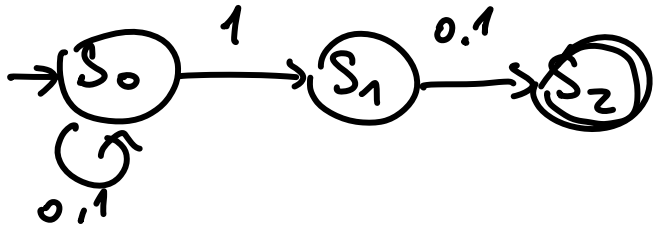
δ	a	b
→ s_0	$\{s_0, s_1\}$	$\{s_0\}$
s_1	$\emptyset$	$\{s_2\}$
◎ s_2	$\emptyset$	$\emptyset$

δ^d	a	b
→ $\{s_0\}$	$\{s_0, s_1\}$ ✓	$\{s_0\}$ ✓
$\{s_0, s_1\}$	$\{s_0, s_1\}$ ✓	$\{s_0, s_2\}$ ✓
◎ $\{s_0, s_2\}$	$\{s_0, s_1\}$ ✓	$\{s_0\}$ ✓

A_1^d

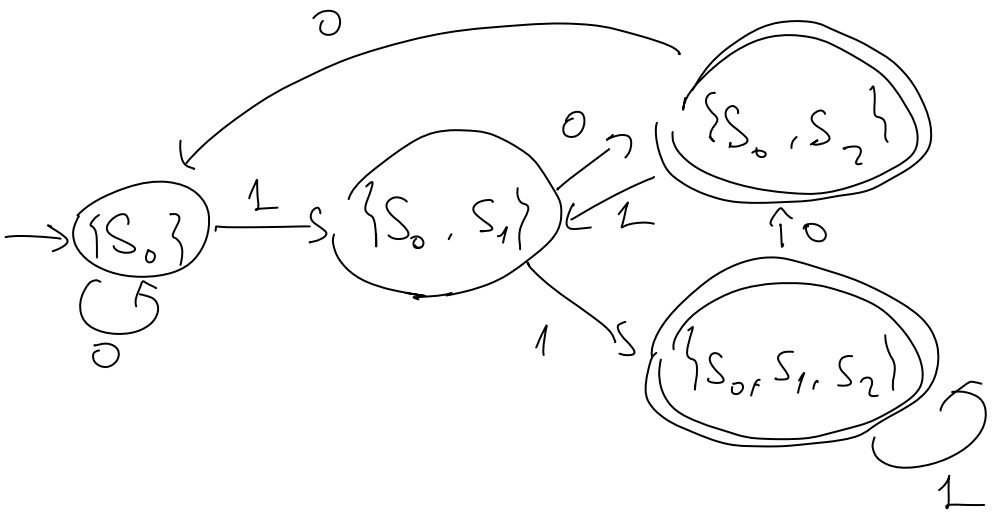


b) $L = \{ w \in \{0,1\}^* \mid w = xLz,\ x \in \{0,1\}^*,\ y \in \{0,1\} \}$

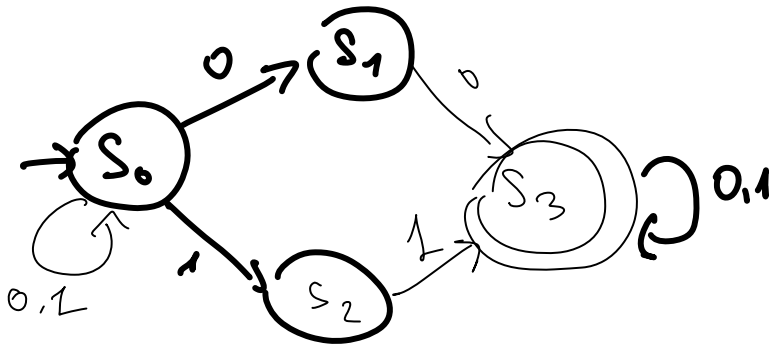


δ	0	1
→ s_0	$\{s_0\}$	$\{s_0, s_1\}$
s_1	$\{s_2\}$	$\{s_2\}$
◎ s_2	$\varnothing$	$\varnothing$

δ^c	0	1
→ $\{s_0\}$	$\{s_0\}$ ✓	$\{s_0, s_1\}$ ✓
$\{s_0, s_1\}$	$\{s_0, s_2\}$ ✓	$\{s_0, s_1, s_2\}$
◎ $\{s_0, s_2\}$	$\{s_0\}$ ✓	$\{s_0, s_1\}$ ✓
◎ $\{s_0, s_1, s_2\}$	$\{s_0, s_2\}$ ✓	$\{s_0, s_1, s_2\}$ ✓

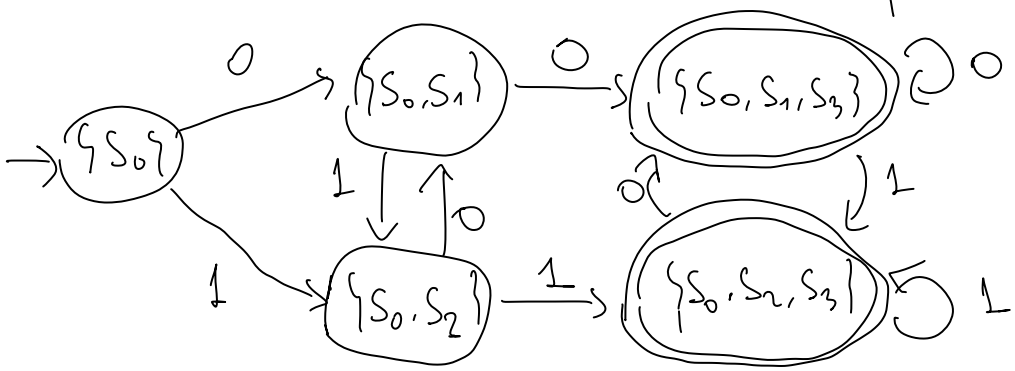


c) $L_3 = \{ w \in \{0,1\}^* \mid w = x00y \vee w = x11y, x,y \in \{0,1\}^* \}$



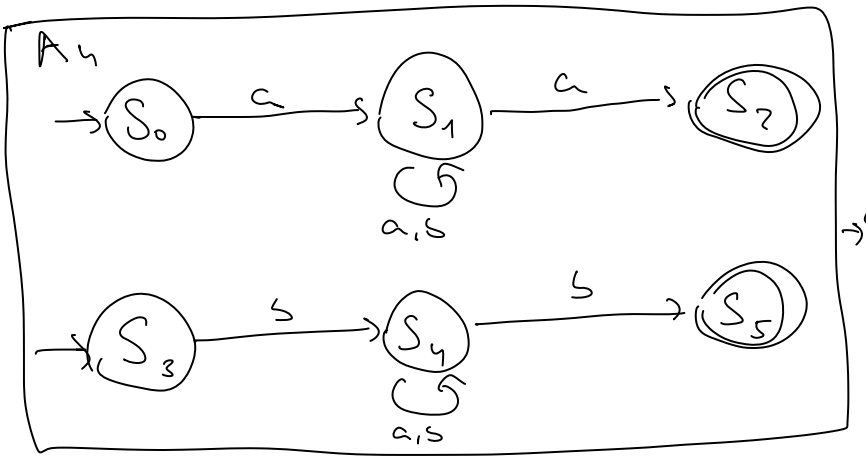
δ	0	1
$\rightarrow s_0$	$\{s_0, s_1\}$	$\{s_0, s_2\}$
s_1	$\{s_3\}$	$\emptyset$
s_2	$\emptyset$	$\{s_3\}$
◎ s_3	$\{s_3\}$	$\{s_3\}$

δ^d	0	1
$\rightarrow \{s_0\}$	$\{s_0, s_1\}$ ✓	$\{s_0, s_2\}$ ✓
$\{s_0, s_1\}$	$\{s_0, s_1, s_3\}$	$\{s_0, s_2\}$ ✓
$\{s_0, s_2\}$	$\{s_0, s_1\}$ ✓	$\{s_0, s_2, s_3\}$ ✓
◎ $\{s_0, s_1, s_3\}$	$\{s_0, s_1, s_3\}$ ✓	$\{s_0, s_2, s_3\}$ ✓
◎ $\{s_0, s_2, s_3\}$	$\{s_0, s_1, s_3\}$ ✓	$\{s_0, s_2, s_3\}$ ✓



d) $L_4 = \{ w \in \{a,b\}^* \mid w = xyx,\ x \in X,\ y \in X^* \}$

Sind die Wörter "a" und "b" erlaubt?



δ^d	a	b
→ $\{S_0, S_3\}$	$\{S_1\}$ ✓	$\{S_4\}$ ✓
$\{S_1\}$	$\{S_1, S_2\}$	$\{S_1\}$ ✓
$\{S_4\}$	$\{S_4\}$ ✓	$\{S_4, S_5\}$ ✓
◎ $\{S_1, S_2\}$	$\{S_1, S_2\}$ ✓	$\{S_1\}$ ✓
◎ $\{S_4, S_5\}$	$\{S_4\}$ ✓	$\{S_4, S_5\}$ ✓

